

Vanika Sok

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Education

Lafayette College

Bachelor of Arts in Mathematics
Bachelor of Arts in Economics (Honors)

Easton, PA, USA
Aug 2022 – May 2026

Graduated Cum Laude with Honors in Economics

Primary advisor: Dr. David Stifel
Committee: Dr. Susan L. Averett, Dr. Daniel Turek

Relevant Coursework

Real Analysis, Discrete Structure, Probability, Mathematical Statistics, Theoretical Math, Abstract Algebra, Advanced Regression Analysis, Linear Algebra, Stochastic Processes
Fundamentals of Econometrics, Microeconomics, Macroeconomics, Financial Market, Development Economics, Industrial Organization, Independent Study on Advanced Micro Development Economics, Women and the Economy, Thesis

Extracurricular

Orientation Leader and First-Year Mentor; Mentor for Lafayette Initiative for Malagasy Education (LIME)
Lafayette Investment Club, International Student Association, Women in Computing, Women in Math

Teaching Experience

Teaching Assistant, Lafayette College

ECON 253: Econometrics; Professor Matthew Larsen Fall 2025
PHYS 131: Mechanics; Professor Paul Stonaha Spring 2025
ECON 251: Intermediate Microeconomics; Professor Sayorn Chin Spring 2024

Honors and Awards

Omicron Delta Epsilon (ODE), the International Honor Society for Economics Apr 2025
Founding Chapter Leader, National Student Data Corps (NSDC), Lafayette College–Northeast Sep 2023
Big Data Innovation Hub

Best Overall Hackathon Award

Feb 2023

Best overall hackathon award from Lafayette College's Association for Computer Machinery for a project on a mental health website featuring a chatbox, games, and videos designed to alleviate stress.

SHE-CAN Scholar

Jan 2020 – Present

One of four scholars selected nationally for the SHE-CAN scholarship, based on academic excellence, leadership, project management, and public speaking.

Professional Experience

United Nations Office for Project Services (UNOPS)

New York, NY

Digital & Innovation Strategy Intern (AI, GIS & IM)

May 2026 – Present

Support Compass, UNOPS's geospatial innovation and analytics facility within the Information Management and Analytics Team (IMAT), applying satellite data and AI to crisis response in conflict-affected contexts. Document interpretability of machine learning terrain models identifying explosive ordnance risks; produce communication materials on AI tools (CARE, ORBIT); prepare damage assessments and environmental mapping from geospatial observation data; catalog high-resolution satellite imagery for 3D modeling and rubble estimation in reconstruction zones.

Research Experience

University of Utah, Office of Undergraduate Research

Salt Lake City, UT

Researcher, Dept. of Economics

May 2025 – Aug 2025

Developed research design from the ground up—synthesizing literature across unemployment, childcare markets, the care economy, and public goods provision to frame causal questions and the identification strategy. Engineered end-to-end data pipelines via webscraping and RESTful APIs to merge 15+ years of panel data from the Public Library Survey, BLS, and Business Dynamics Statistics using Python, R, and Stata. Applied OLS, fixed-effects, and instrumental variable models to produce the first causal estimates of unemployment's effect on U.S. public library usage. Co-wrote manuscript through submission; one paper under review, a second in progress.

Applied two-way fixed-effects models to NLSY97 and O*NET data to quantify a statistically significant wage premium for morally compromising occupations, with premium magnitude increasing with age and education. Presented findings at Lafayette College's Poster Symposium, demonstrating how markets price moral considerations and offering insights with implications for workforce management, compensation strategies, and organizational decision-making.

Analyzed how crop diversification affects household welfare in Nigeria, leveraging World Bank panel survey data. Applied econometric models, data wrangling, and visualization in Excel and Stata to quantify the relationship between agricultural diversity and economic well-being. Produced insights on how diversification strategies can strengthen household resilience and inform development policy.

Research

“Does Women’s Empowerment Improve Child Welfare? New Evidence from Cambodia”

Abstract Women’s empowerment in developing countries remains understudied, and in particular its impact on child welfare. I examine how women’s intra-household bargaining power – proxied by women’s land ownership, education, perception about justified wife-beating, and household decision-making – affects child welfare indicators. Using Demographic and Health Survey (DHS) data from Cambodia from 2005 to 2022, I estimate how women’s empowerment affects child school enrollment status, immunization rates, and child anthropometric health outcomes. I also explore whether these impacts differ by the gender of the child, and potentially reduce any gender-based inequality in child welfare. In order to reduce the bias in the estimators of these effects, I control for household level characteristics and region and time fixed effects. The findings will contribute to the literature on gender and development by providing new evidence from Cambodia and offering insights for policies targeting child welfare.

“Unemployment and Public Library Usage in the United States: Evidence from Before and During COVID” Co-authored with Sarah Small (University of New England), Rachel Ernst (University of Utah), and Laura Beltran Figueroa (Rutgers University). *Under review.*

Abstract Scholars and policymakers have long argued that public libraries offer essential community resources during times of economic strife. However, to our knowledge, most of this evidence has been anecdotal or correlational. Using Public Library Survey, Bureau of Labor Statistics, and Business Dynamic Statistics Datasets, we present causal estimates of the effect of regional unemployment on local public library usage. Evidence suggests higher unemployment consistently leads to increased library visits, as well as public internet computer use in both the pre-COVID and COVID-era periods. However, the relationship between unemployment and specific library use varies: programming and attendance show limited pre-COVID evidence but positive COVID-era associations, while circulation outcomes are more mixed and depend on the type of material. These findings suggest that libraries’ countercyclical role depends not only on community need, but also on the specific service channel through which patrons access library resources.

“Do Public Libraries Fill Care Gaps? Regional Analyses of Children’s Library Programming, Childcare Markets, and Parenthood” coauthored with Sarah Small and Rachel Ernst. *Under review.*

Abstract Public libraries are regularly relied upon for community resources beyond just books. In this paper, we ask: do public libraries help fill care gaps? Using data from the United States Public Library Surveys and the National Database on Childcare Prices, we construct a panel of public libraries from 2008 to 2022 matched with corresponding demographic and childcare pricing information in their respective counties. Using these data, we estimate how family demographics and care costs relate to attendance in children’s programming and circulation of children’s books in public libraries. Ultimately, we find that in counties with more single mothers, higher home-based childcare prices, and higher rates of family-level poverty, libraries have higher attendance in young children’s programming activities and greater children’s book circulation. We test the robustness of our results using several instrumental variable analyses and falsification tests. Ultimately, we suggest that libraries serve as important public spaces of social reproduction.

Works in Progress

“Further Evidence on Pollution, Crime, and Anti-Social Behavior” coauthored with Sayorn Chin (Lafayette College) and Salvador Lurbé (Lead Data Scientist at Mountain Data Group)

Abstract Recent literature has identified an association between short-term pollution exposure and crime in the US and in the United Kingdom (UK). In this paper, we expand our understanding of the effects of pollution on crime by looking at the association between pollution, specifically at Particulate Matter (PM) and Ozone (O3), and counts of aggregated crime types and Anti-social Behavior (ASB) in the UK. We primarily focus our analysis on ASB, as the literature has identified costs associated with them in the UK, but has overlooked its association with pollution. Using a fixed effects model, we found an association between pollution and some crime types, especially those that are economically motivated. We find weaker evidence of an association between pollution and offenses associated with aggressive and violent behavior (including ASB), and we discuss potential mechanisms in the context of the rational choice crime model. We conclude that one potential mechanism could be a decrease in the utility of the non-punishable alternative activity, or a decrease of the offender’s value of the future costs associated with being caught.

Research
Presentation

Poster Symposiums and Presentations

Presented research poster “Unemployment and Public Library Usage in the United States” at Lafayette College Symposium (Fall 2025) and Lafayette Bicentennial Celebration.

Presented research poster “The Compensation of Conscience” at Lafayette College Poster Symposium (Spring 2025).

Presented research poster “Unemployment and Public Library Usage in the United States” at University of Utah Undergraduate Research Symposium (Summer 2025).

Shared findings on childcare markets and public library programming at University of Utah Summer Research Symposium (2025).

Research
Assistant

“The Compensation of Conscience”, Sayorn Chin

This paper has been submitted to the *Journal of Economic Behavior and Organization*

Abstract Markets are often praised for their efficiency, but they can also incentivize individuals to act against their moral convictions. This study quantifies the “price of conscience” in labor markets, investigating whether jobs requiring moral compromise offer higher wages. Using data from the National Longitudinal Survey of Youth 1997 (NLSY97) and Occupational Information Network (O*NET), we conduct a longitudinal fixed-effects analysis to examine how conscience-related job characteristics influence wages. Our findings suggest that occupations demanding greater moral compromise provide higher wages, and this premium increases with age and education. These results extend prior experimental evidence on morals and markets into real-world labor market dynamics.

Term
Projects

Department of Mathematics, Lafayette College

Probabilistic Forecast Calibration and Model Validation Study of Major League Baseball
Regression-Based Time-Series Enrollment Forecasting for Lafayette Spring 2026 Math Courses
Proportional-Odds Modeling of Conspiracy Theory Beliefs Using Information-Criterion-Based Selection

Department of Economics, Lafayette College

Analyzing JetBlue’s Strategy: From Levitt’s Theory to Limited Information and Pricing
Incomplete Contracts and Market Breakdown in High-Risk Homeowners Insurance
Does Rigorous High School Exposure to STEM Education Improve Long-Term Employment Outcomes?
Lafayette Dining Optimization: Utility and Risk Modeling

Professional
Certifications

UN & UNOPS

Prevention of Sexual Exploitation and Abuse by UN Personnel
Diversity, Equity and Inclusion at UNOPS

New York, NY
Jul 2026
Jul 2026

IBM

Data Science Specialization

Online
May 2024

Training the Street

New York, NY

Skills

Programming

Stata, Python, R, SQL

Methods

Causal Inference: RCTs, Diff-in-Diff, Regression Discontinuity, Instrumental Variables, Fixed Effects.
Geospatial & Data: GIS, Spatial Analysis, Data Visualization, Web Scraping, APIs.

Software

Git/GitHub, ArcGIS/QGIS, $\LaTeX$, Google Cloud (BigQuery, Vertex AI), Tableau, Microsoft Excel.

Languages

English (fluent), Khmer (native), Chinese (limited working proficiency), French (beginner)

Conferences

Annual Bank Conference on Development Economics (ABCDE), 2026

Washington, DC

Jobs for the Next Generation. Attended the World Bank's flagship development economics conference, hosted at World Bank Headquarters. Engaged in Q&A and panel discussions with leading economists and policymakers, including Nobel laureate Daron Acemoglu.

The National Conference on Undergraduate Research (NCUR), 2026

Richmond, VA

Does Women's Empowerment Improve Child Welfare? New Evidence from Cambodia. Oral presenter and session moderator for undergraduate student presenters.

Economic Scholars Program, Federal Reserve Bank of Cleveland, 2026

Cleveland, OH

Federal Reserve-hosted economic research conference for students to share research, network with professionals, and explore Fed career pathways.

UN International Day of Women and Girls in Science Celebration, 2024

New York City, NY

Represented Lafayette College's Inclusive STEM Education initiatives at the United Nations headquarters. Participated in high-level UN sessions on gender equity in STEM education and workforce participation. Applied policy insights to strengthen LIME mentoring initiatives, Women in Math, and Women in Computing outreach programming on Lafayette campus.

Grace Hopper Celebration (GHC), 2023

Orlando, FL

Represented Lafayette College's Women in Computing at the Grace Hopper Celebration, the world's largest conference for women in technology. Participated in technical sessions, industry panels, and the career fair; engaged with recruiters, engineers, and researchers on emerging trends in AI, data science, cybersecurity, and quantitative computing.

SHE-CAN Leadership Summit 2024 and 2026

San Francisco, CA

Annual, weekend long, in-person summit in the San Francisco Bay Area. Summit Mission Statement: The "Perpetual Power of One" is the realization that individual actions and contributions, no matter how small they may seem at the outset, can create a ripple effect that continues over time and inspires others to make their own contributions.

Leadership Summit, 2023

San Francisco, CA

Selected to attend the 2023 SHE-CAN Leadership Summit, an annual weekend-long event in the San Francisco Bay Area, focused on how leadership can drive sustainability, conscious capitalism, social justice, and the empowerment of women.

Leadership
and Volunteering**Lafayette Data Community**

Easton, PA

Board Member

Apr 2023 – Present

Coordinate campus-wide data science programming, host technical workshops, and invite speakers on AI, quantum computing, and computational sustainability. Initiate and lead Inspire-a-Thon with Hanson Center, a campus-wide AI-focused innovation challenge featuring multidisciplinary project tracks in policy, technology, and creative expression.

Lafayette Extended Orientation (LEO) Program

Easton, PA

LEO Leader and First-Year Student Mentor

Apr 2025 – Present

Provides support and mentorship to new students Class of 2029 during their first year as they navigate their new intellectual community both socially and academically.

Lafayette Initiative for Malagasy Education (LIME)*Math Tutor*

Madagascar

Dec 2024 – Present

Selected for a 1.5-year commitment with the LIME; traveled to Antananarivo, Madagascar (Winter 2025) to mentor high school students, providing academic guidance, leadership training, and support in preparing for competitive university and scholarship applications to the U.S.

Lafayette Intercultural Development*International Orientation Captain*

Easton, PA

May 2023 – May 2024

Led the planning and execution of a week-long International Orientation program serving students from 52 countries in the Classes of 2027 and 2028.

Cambodian Student Association in the United States*Director of Finance*

Washington, DC (Remote)

Sep 2023 – May 2025

Managed all financial operations, securing grants from the Cambodian Embassy and extending the student budget by 125 percent. Oversaw fund allocation across departments to maximize ROI, maintained full financial records, and collaborated with the Board of Directors to authorize and disburse funds in compliance with CASUSA's constitution, supporting the association's mission to build a welcoming community for Cambodian students in the U.S.

SHE-CAN Make A Change Project: Higher Education Fair*Founder*

Siem Reap, Cambodia

Mar – Jun 2024

Brought the first-ever Higher Education Fair to live at a Rural High School in Siem Reap province. Drawn over 300 students participation from 12 different high school across Cambodia and invited four incredible speakers. All this made possible due to the help of my younger sister, and a team of six brilliant volunteers whom we recruited from 200+ student who filled out our volunteer application form.

Center of Khmer Studies and American Corner*1st and 2nd Book Fair Team Leader and Speaker*

Siem Reap, Cambodia

Apr – Jul 2022 and May – Jun 2023

Planned and executed 3-month project to organize inaugural book fair in Siem Reap. Sent invitations to 20 high schools, universities, and other educational institutions, driving 5,000 in-person attendees. Collaborated with 25 publishers, libraries, bookstores, universities, and educational institutions to display their products, securing donations, increasing event attendance 525 percent over projections. Supervised Booth and Registration teams and emceed 5000 person event. After the great success of the first book fair, our team returned in 2023 with more major exhibitors from across the country.

United States Agency for International Development*Student Representative*

Siem Reap, Cambodia

Nov 2021

Led group discussions on career development and education challenges in rural areas Cambodia, delivering public speeches and fostering group discussions on 5-year plans to develop Cambodia.

Cambodia Women's Crisis Center*Community Service Volunteer*

Siem Reap, Cambodia

Aug – Oct 2021

Organized events for 35 families in my community to raise awareness of domestic violence and sex trafficking. Organized events with the center to raise awareness and educate community and incoming tourists. Directed underprivileged households to the hotlines number to reduce impact of domestic assault across Cambodia.

References**David Stifel (Advisor)**

Professor, Department of Economics
Department Head
Lafayette College
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Jeff Liebner

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Mathematical Sciences
Assistant Department Head
Lafayette College
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Sayorn Chin

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